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BE-B

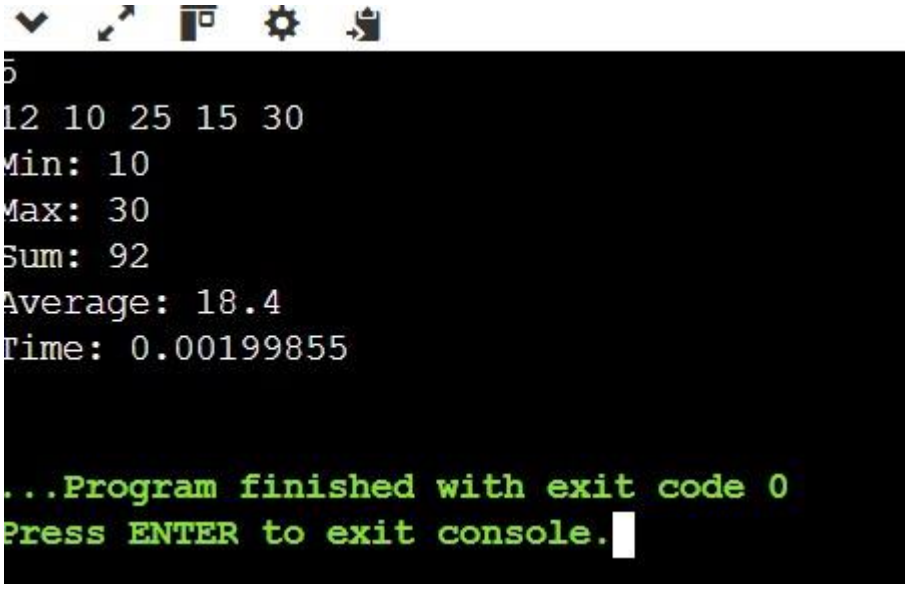
B22069

Sub-HPC

Assignment 3

Implement Min, Max, Sum and Average operations using Parallel Reduction

Code:

```
#include <iostream>

#include <vector>

#include <omp.h>

#include <limits> using

namespace std;

int main() {

    int n;     cin >> n;

vector<int> arr(n);     for

(int i = 0; i < n; i++) {

cin >> arr[i];

    }


    int sum = 0;     int min_val =

numeric_limits<int>::max();     int max_val =

numeric_limits<int>::min();     double start =

omp_get_wtime();



    #pragma omp parallel for reduction(+:sum) reduction(min:min_val)
reduction(max:max_val)

    for (int i = 0; i < n; i++) {         sum +=
arr[i];         if (arr[i] < min_val) min_val =
arr[i];
```

```
        if (arr[i] > max_val) max_val = arr[i];
    }

    double avg = (double)sum / n;
    double end = omp_get_wtime();
    cout << "Min: " << min_val << endl;
    cout << "Max: " << max_val << endl;
    cout << "Sum: " << sum << endl;
    cout << "Average: " << avg << endl;
    cout << "Time: " << end - start << endl;
    return 0;
}
```